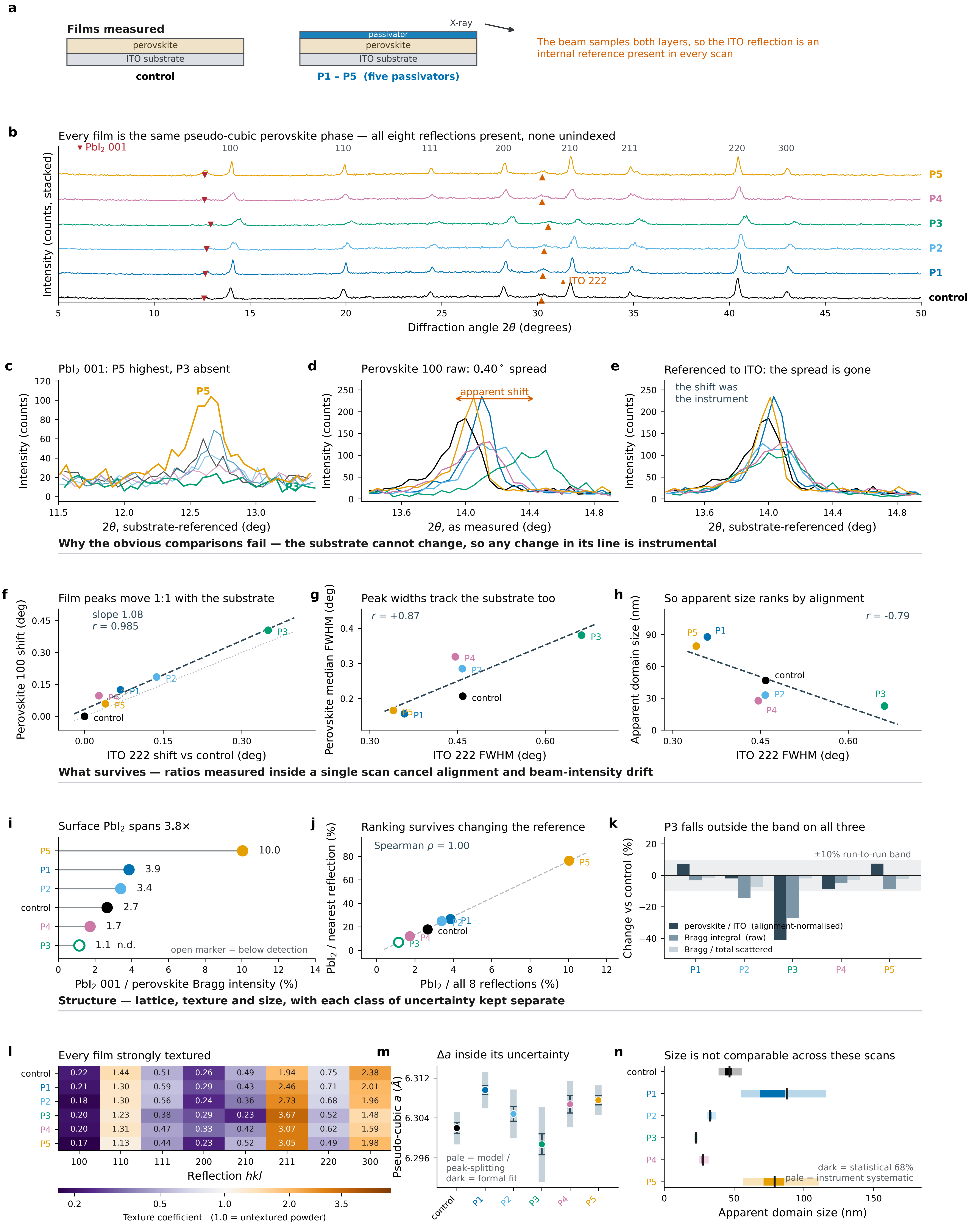
Five surface passivators on perovskite films, screened by X-ray diffraction

One scan per film (n = 1, no repeats). Identical protocol throughout: Cu Kα λ = 1.54184 Å, 5–50° 2θ, 0.05° step, 0.3 s per step, Bragg–Brentano geometry, monochromator on.



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Phase identity — same pseudo-cubic cell	VALID	all 8 reflections index in all six films
PbI_2 / perovskite Bragg ratio	VALID	within-scan ratio; ranking robust (ρ = 1.00)
Comparative crystallinity indices	VALID	protocol identical across all six scans
Texture coefficients	VALID	fixed convention; substrate reflection excluded
Lattice parameter Δa	NOT COMPARABLE	peak shifts track the substrate 1:1 (r = 0.985)
Crystallite size ΔD	NOT COMPARABLE	peak widths track the substrate (r = +0.87)
Absolute degree of crystallinity	BOUNDED RANGE ONLY	amorphous halo degenerate with background
Microstrain	SUPPRESSED	Williamson–Hall slope negative in 5 of 6